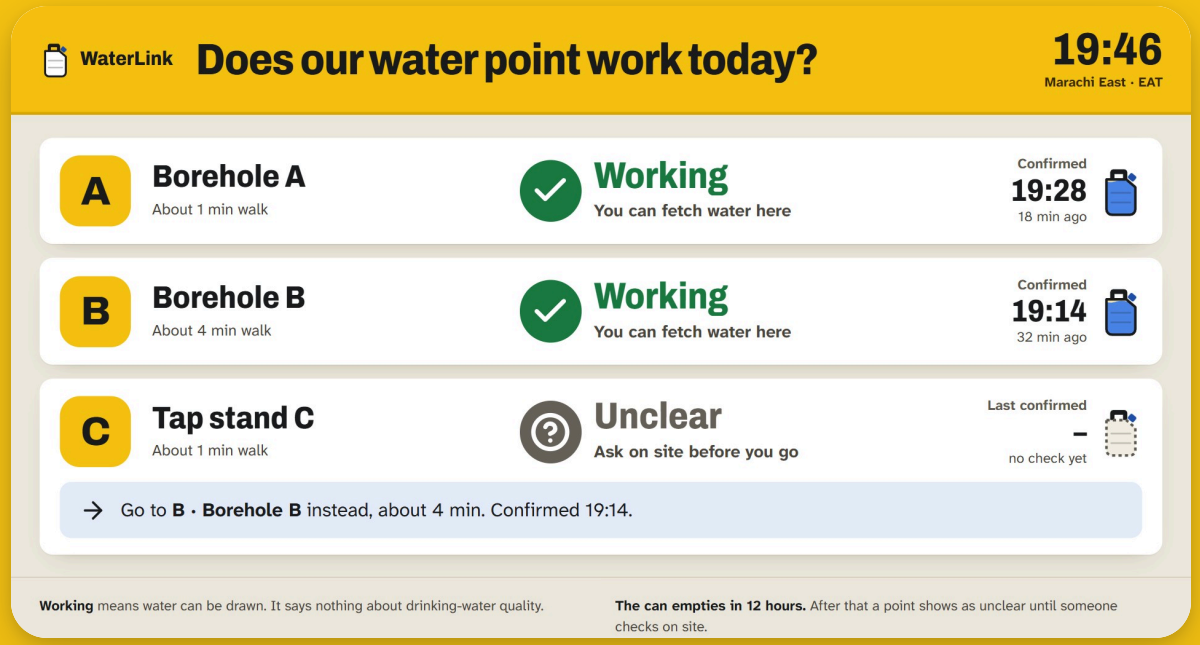
WaterLink

Does our water point work today?

The last confirmed state of every water point, and a repair that only counts once it is tested on site.



Zero/One Hackathon · WU Vienna · 22 October 2026 · Track: Social Impact
Recep Baş · Osman Türkmen

A broken water point looks like a working one until you get there.

Most rural families fetch water at a hand pump or a protected well. When it breaks, the news travels slowly: the walk is wasted, the repair report disappears, and nobody is clearly responsible.

Up to one in three hand pumps does not work at any one time across Africa.

UNICEF Office of Innovation, “Smart Pumps”

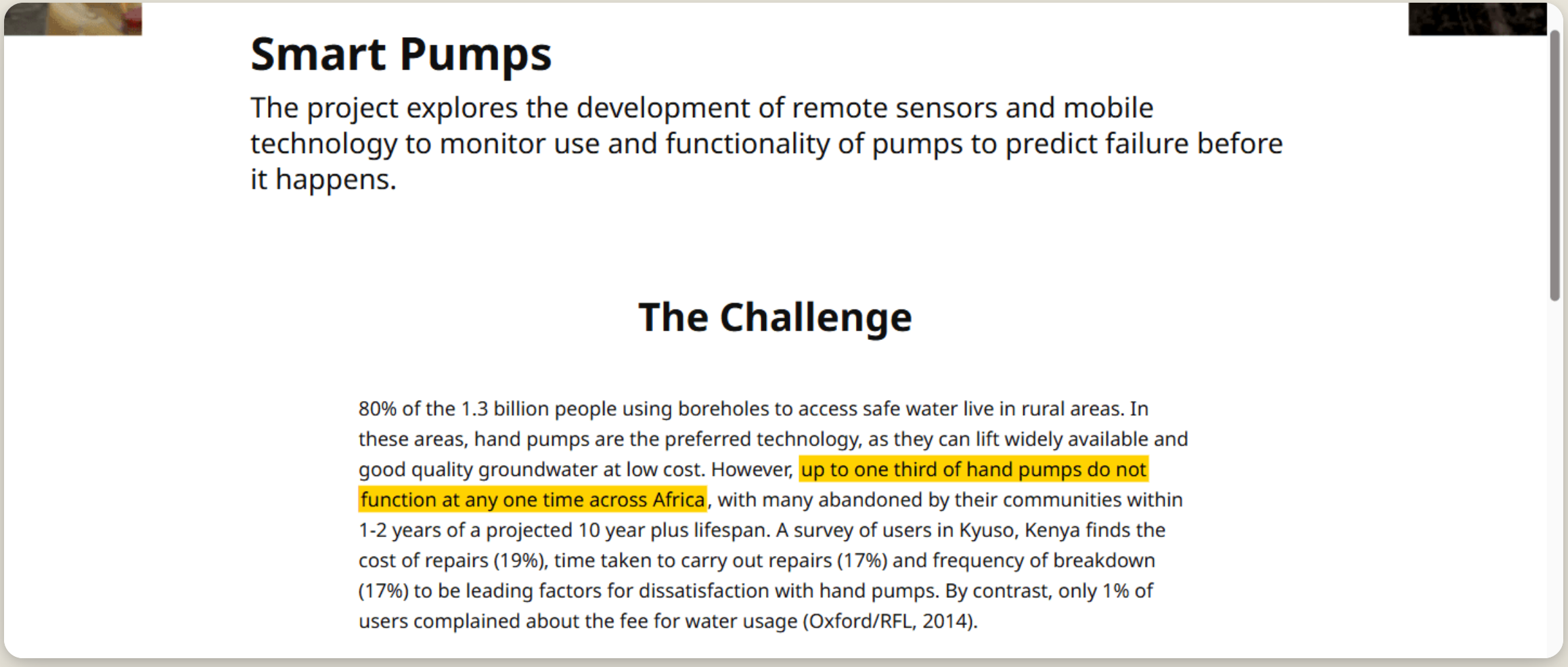
Even with a repair contract, people can be without clean water for at least 2–3 days.

UNICEF, “Smart Pumps”: repairs are due 48–72 hours after a sensor reports a failure

In Kyuso, Kenya, the cost of repairs (19%), the time they take (17%) and how often pumps break (17%) top the list of complaints about hand pumps.

UNICEF, citing a user survey (Oxford/RFL, 2014)

UNICEF: up to one third of hand pumps do not function at any one time



Smart Pumps

The project explores the development of remote sensors and mobile technology to monitor use and functionality of pumps to predict failure before it happens.

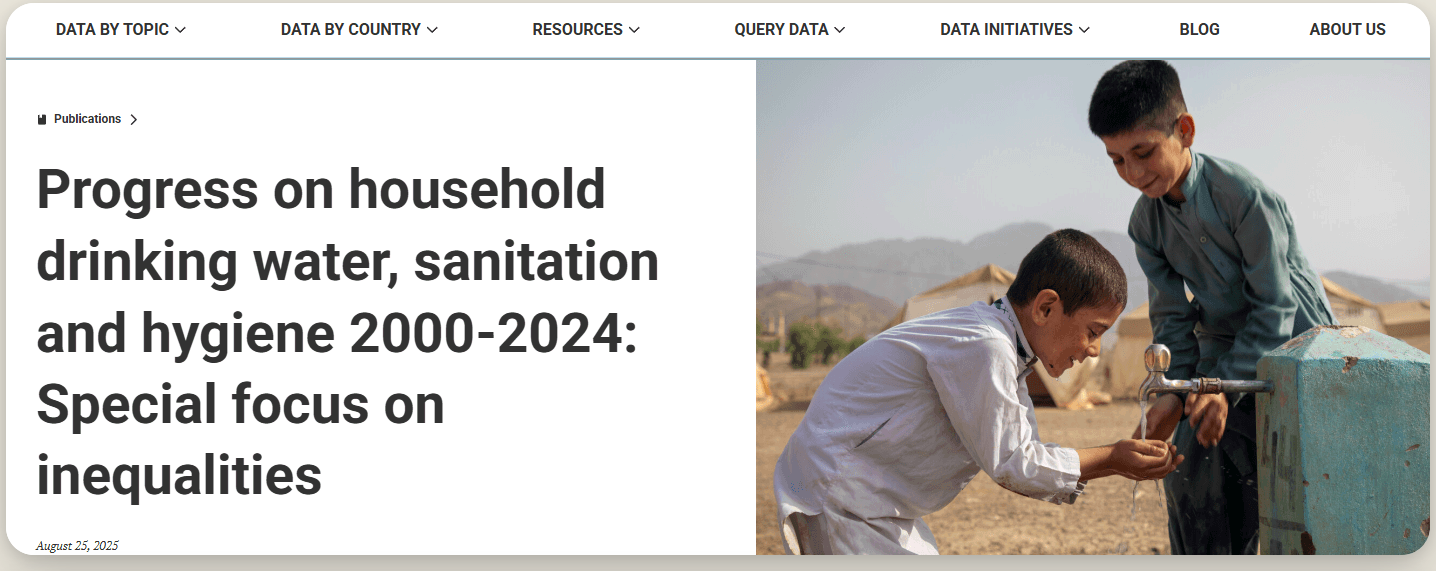
The Challenge

80% of the 1.3 billion people using boreholes to access safe water live in rural areas. In these areas, hand pumps are the preferred technology, as they can lift widely available and good quality groundwater at low cost. However, up to one third of hand pumps do not function at any one time across Africa, with many abandoned by their communities within 1-2 years of a projected 10 year plus lifespan. A survey of users in Kyuso, Kenya finds the cost of repairs (19%), time taken to carry out repairs (17%) and frequency of breakdown (17%) to be leading factors for dissatisfaction with hand pumps. By contrast, only 1% of users complained about the fee for water usage (Oxford/RFL, 2014).

Screenshot of unicef.org/innovation/smart-pumps (UNICEF Office of Innovation), captured 22 October 2026. The yellow highlight is ours; the unmarked capture is in the attached files.

WaterLink

WHO/UNICEF, 2025: 2.1 billion people still lack safe drinking water at home



1 in 4 people around the world lack safely managed drinking water

Between 2015 and 2024, 961million people gained access to safely managed drinking water services, increasing global coverage from 68% to 74%. Rural coverage rose from 50% to 60%, while urban coverage remained steady at 83%. Despite this progress, 2.1 billion people still lack safely managed drinking water, 1.4 billion rely on basic services, 287 million on limited services, 302 million use unimproved sources, and 106 million continue to drink surface water. The number of people using surface water has decreased by 61

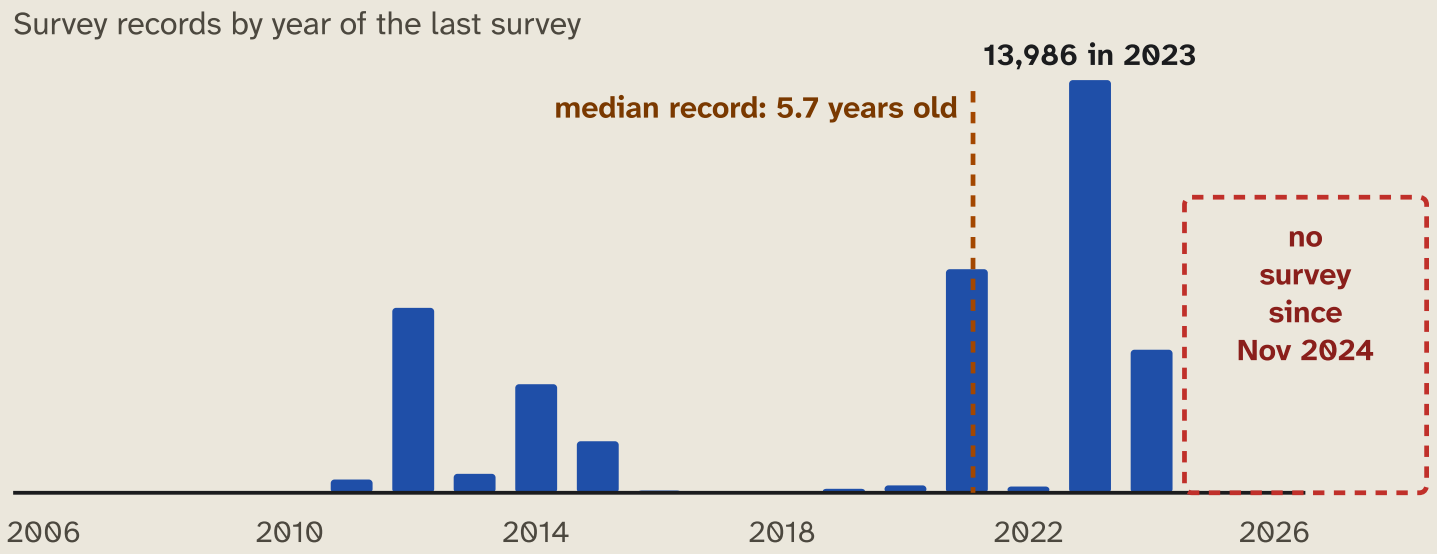
1 in 4 people worldwide lack safely managed drinking water.

Rural coverage is 60%, urban 83%. 106 million people still drink surface water.

"Progress on household drinking water, sanitation and hygiene 2000–2024: Special focus on inequalities", WHO/UNICEF Joint Monitoring Programme, published 25 August 2025. Page data.unicef.org/resources/jmp-report-2025, captured 22 October 2026. Highlight is ours.

WaterLink

The public data cannot tell a family whether the pump works today.



39,959 survey records for Kenya in the Water Point Data Exchange.

The newest was taken on **30 November 2024**. None is from the last 12 months.

The median record is **5.7 years** old; two in three are older than 3 years.

At their last survey **21%** did not work (**27%** in Busia County).

Water Point Data Exchange (WPdx), data.waterpointdata.org, Kenya, retrieved 22 Oct 2026; our calculation (status taken from the visit-day water flag where a dataset's status is constant). A survey record is a visit, not a guarantee of coverage.

WaterLink

The last confirmed state before the walk, and a repair that is someone’s job

1 **A public board.**

Each water point shows its state, who confirmed it and for how long it holds. After 12 hours it says “unclear”, never an old green. It works on a screen at the shop, on a phone, and as a printed sheet.

2 **A repair that counts.**

An outage report becomes a job with a named person and visible blockers (parts, money, staff). “Repaired” is not “working”: only a test on site turns the point green.

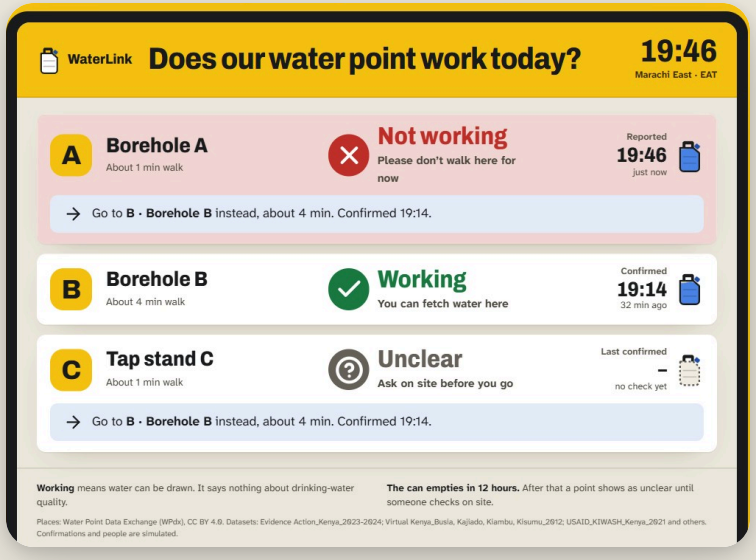
3 **A reliable alternative.**

If a point is down, the board names the nearest one, but only if it was freshly confirmed and its route and access were checked.



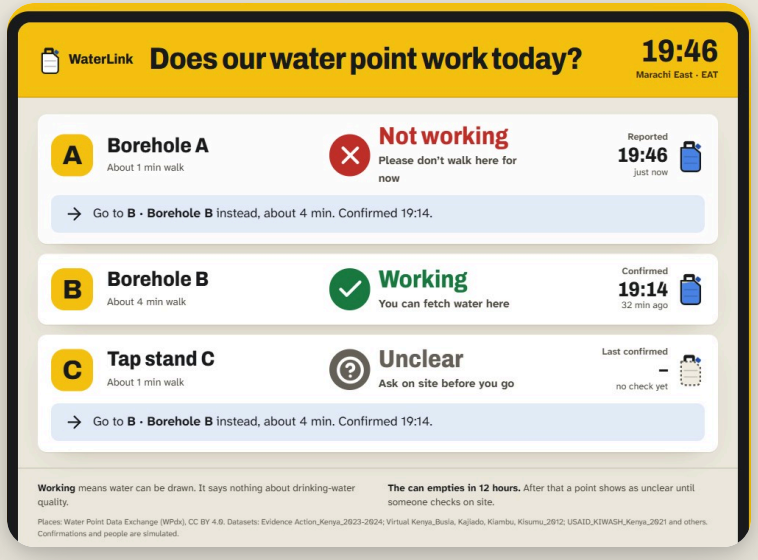
The same board in Kiswahili (draft translation, labelled on the board), with the QR code to the SMS view. Also available as phone view and printed sheet.

One outage, from the first report to a repair that counts



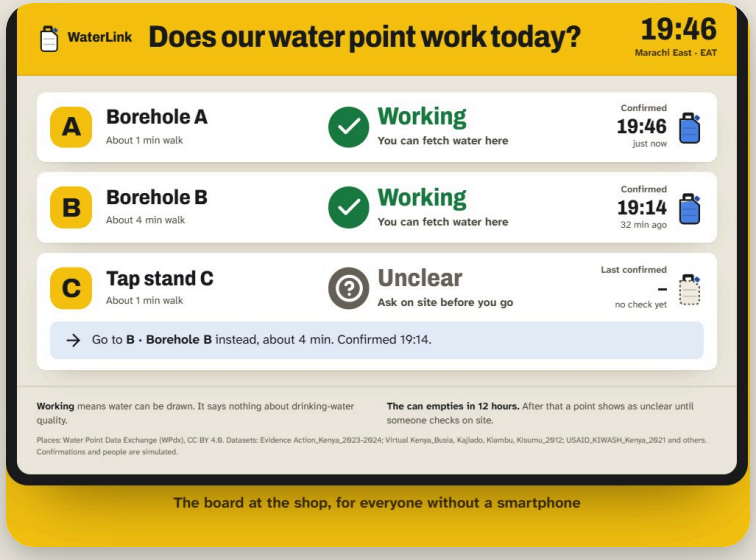
Report

The operator reports no water at A. The board turns A red at once and names B. A technician gets the job.



Repaired is not working

The technician reports the repair. The board stays red until someone tests the pump on site.



Tested on site

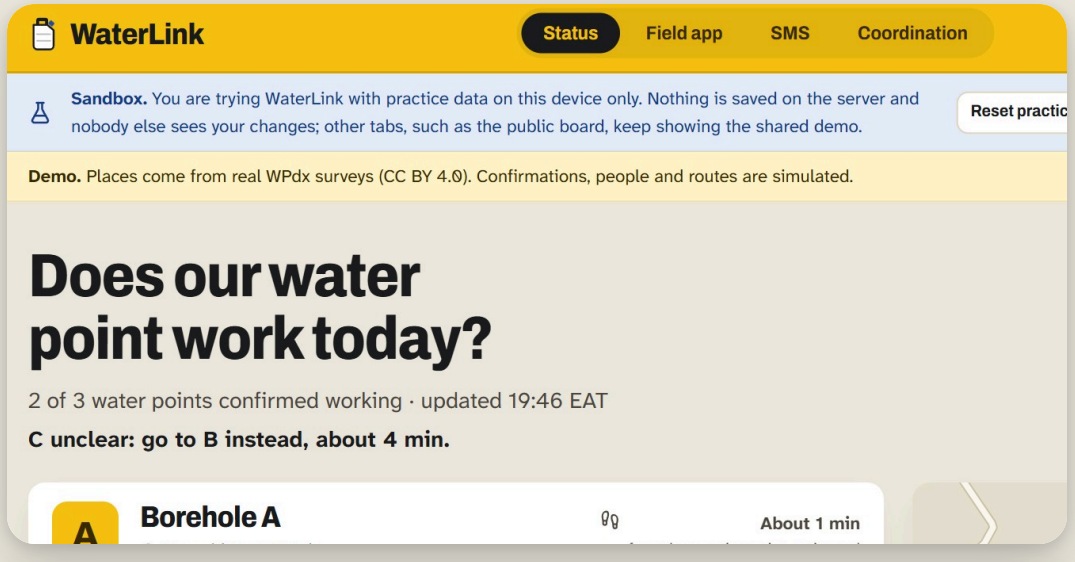
The operator pumps 20 litres and confirms. Only now A is green again, with the time of the check.

Screens from the practice mode of the prototype. People, confirmations and routes are simulated.

WaterLink

A working prototype with the important features, live today

- **Public board and household view:** state, time of the check, validity; English and Kiswahili (draft); printable sheet.
- **Report → job → progress and blockers → repair → test on site,** with history; roles enforced on the server.
- **Uncertainty is shown, not hidden:** stale, conflicting, offline and sensor-hint states never look working.
- **Alternatives only when verified:** route and access checks, opening hours, adding a water point by a tap on the map.
- **SMS simulation** and a replay of real 2014 smart-handpump data (Kwale) as a sensor hint. **Practice mode** for anyone, no login.
- **Checked:** 85 unit tests and 37 browser tests (repair flow, offline, accessibility, phone width, print) on every push.





What is real, what is simulated, what is not connected

Real

42 places in Marachi East, Busia, from 60 survey records of the Water Point Data Exchange (CC BY 4.0); repeated surveys of one place are merged.

Roads and places from OpenStreetMap.

Weekly pump volumes from Kwale, Kenya, 2014 (openwashdata, CC BY 4.0).

All rules, the server and the tests.

Simulated, and labelled so

People: the operator, the technician and coordination.

Confirmations, reports and repairs.

Walking routes and opening hours.

The state of the places: survey status is historical and never shown as current.

Not connected yet

An SMS gateway for real messages.

Accounts per water point (today: one shared demo code).

Live sensors, payments, an AI service.

A pilot partner. We make no impact claims.

“Working” means water can be drawn. It says nothing about drinking-water quality, and every board says so.

Numbers behind our assumptions

Assumption	Evidence	Source	Status
Families cannot learn today's state from existing data	Newest of 39,959 Kenyan survey records: 30 Nov 2024. Median age 5.7 years, 67% older than 3 years.	WPdx, own calculation	**Supported**
Many water points are broken at any time	Up to 1 in 3 hand pumps in Africa. 21% of Kenyan records non-functional at the last survey (27% in Busia).	UNICEF; WPdx	**Supported**
Waiting for repairs is what hurts	Repair time (17%), breakdown frequency (17%), repair cost (19%) lead the complaints in Kyuso, Kenya. At least 2–3 days without clean water.	UNICEF (Oxford/RFL 2014)	**Supported**
An organised repair service works	Handpump downtime in Kwale and Kitui fell from 27 days to under 3 with sensors and a maintenance service.	WASREB 2014 via REF 2021; SDSN TReNDS 2018	**Other project**
Households without smartphones can be reached	48.6% of rural and 64.6% of urban people in Kenya own a mobile phone, so the board and a short SMS flow.	Kenya Housing Survey 2023/24 (CA, KNBS)	**Partly**
Operators confirm often enough; 12 hours is the right validity	No evidence yet. 12 hours is a demo parameter.	–	**Open**
Someone will pay	No evidence yet.	–	**Open**

What we have not proven, and how we will measure it

Fresh confirmations

Share of water points with a confirmation younger than its validity. Tells us whether operators really confirm.

Time to a repair that counts

Median time from report to claim, and to the test on site. Shows where the days go: information, money, parts or people.

Wasted walks

Households that walked to a broken point, before and after the board is in place.

Understanding

Does anyone read “Working” as “clean water”? The answer must be no.

The plan

One neighbourhood, a handful of water points, one board, confirmations by a named person per point.

A partner who is responsible for the points: a county water office, an NGO, or a maintenance service of the FundiFix kind.

Baseline first. We measure the four numbers before the board goes up and agree the targets afterwards, not before.

We do not claim an impact today. We have a prototype, a data analysis and a plan to find out.

What we learned while building it

1 An old green is worse than no answer.
So every confirmation expires, and an expired point reads “unclear”.

2 Repaired is not working.
Only a test on site, by the operator, turns a point green again.

3 “Working” is not “safe”.
The board says so every time; we make no water-quality claim.

4 A dot on the map is not a recommendation.
An alternative needs a fresh confirmation and a checked route first.

5 Records are not places.
Checking the data showed that 60 survey records are 42 places. We merged them and said so.

6 Say what is unchecked.
The Kiswahili texts are a draft, compared with Swahili sources, and labelled as a draft on the board.

7 Test where it hurts.
A test run against the live site found a clock bug (a device 9 seconds behind the server) that looking at the screen never showed.

8 Do not invent evidence.
No testimonials, no impact figures. What is open is on the page “Numbers behind our assumptions”.

Next: one neighbourhood, one partner, four numbers

1. **Find the partner and the place.**
   A county water office, NGO or maintenance service, and a neighbourhood where the board can hang and someone passes it on the way to water.

2. **Connect what is simulated.**
   A real SMS gateway, an account for every water point and operator, backups and data protection for Kenya.

3. **Measure.**
   Baseline, then the four numbers on the previous page, with the people who use it.

4. **Decide with evidence.**
   Scale up, change or stop, depending on what the numbers say.

IdeaLAB

We would potentially agree to take part in the IdeaLAB if we are selected. Validation and a first conversation with a partner are exactly where we need the programme most.

Try it yourself, no login needed

Practice mode, the whole story
waterlink-demo.waterlink.workers.dev/?view=stage&sandbox=1

Live demo
waterlink-demo.waterlink.workers.dev

Code and documentation
github.com/OsmanTuerkmen/WaterLink (a copy of the source is attached to our e-mail)

Team
Recep Baş (basreci@gmail.com), Osman Türkmen

Scan to play the story on your phone

Appendix: UNICEF for Eastern and Southern Africa names non-functional handpumps

experiencing droughts and floods that threaten quality and quantity of water. With cholera being endemic in the region, any extreme weather – whether droughts or floods – leads to outbreaks of cholera and other water-borne epidemics. Climate change also impacts heavily on sanitation and hygiene as water scarcity and flooding leads to communities abandoning important sanitation and hygiene behaviors such as use of flooded toilets or toilets that use flushing water and even handwashing. Climate changes are also driving displacement (refugees, IDPs).

Finally, poor management of water supply services is affecting their sustainability. As in other regions in the developing world, the region experiences failure of water systems before their life time, has a high proportion of non-functional systems especially handpumps, breakdown rates are high, and cost-recovery is poor.



Screenshot of unicef.org/esa/water-and-environment (UNICEF Eastern and Southern Africa), captured 22 October 2026. The same page: “Over 226 million people in ESA (47 per cent) have no access to at least basic drinking water services.” Highlight is ours.

Appendix: sources and method

UNICEF Office of Innovation, Smart Pumps. unicef.org/innovation/smart-pumps. Hand pump failure, repair times, Kyuso survey (Oxford/RFL 2014). Captured 22 Oct 2026.

WHO/UNICEF JMP, Progress on household drinking water, sanitation and hygiene 2000–2024. data.unicef.org/resources/jmp-report-2025, published 25 Aug 2025. Captured 22 Oct 2026.

UNICEF Eastern and Southern Africa, Water and environment. unicef.org/esa/water-and-environment. Captured 22 Oct 2026.

Water Point Data Exchange (WPdx). data.waterpointdata.org, CC BY 4.0. Kenya, 39,959 records retrieved 22 Oct 2026. Our method: *web/scripts/wpdx-lib.mjs* in the repository; results in *docs/einreichung/belege/wpdx-kenya-zahlen.json*.

WASREB, Performance Review of Kenya's Water Services Sector 2014, quoted in the REF 2021 impact case study on the Smart Handpump project (downtime 27 days to under 3); SDSN TReNDS case study 2018.

openwashdata, waterpumpkwale (CC BY 4.0, doi:10.5281/zenodo.8399886): pump volumes, Kwale 2014. **OpenStreetMap** contributors (ODbL): roads and places.

Communications Authority of Kenya and KNBS, Kenya Housing Survey 2023/24, published 12 Aug 2025: mobile phone ownership 48.6% rural, 64.6% urban (ca.go.ke).

Our own data: the 42 places come from 60 survey records after merging records of different surveys within 50 m; the demo points A, B and C stay separate. The 12-hour validity and all people, confirmations and routes are simulated.